Question ID: 463eec13

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

If $x \neq 0$, which of the following expressions is equivalent to $\frac{\sqrt{16x^4y^8}}{x^3}$?

Answer

A. $8x^2y^4$

B. $4xy^4$

C. $4x^{-2}y^2$

D. $4x^{-1}y^4$

Correct Answer: D

Rationale

Choice D is correct. Taking the square root of an exponential expression halves the exponent, so $\frac{\sqrt{16x^4y^8}}{x^3} = \frac{4x^2y^4}{x^3}$, which further reduces to $\frac{4y^4}{x}$. This can be rewritten as $4x^{-1}y^4$.

Choice A is incorrect and may result from neglecting the denominator of the given expression and from incorrectly calculating the square root of 16. Choice B is incorrect and may result from rewriting $\frac{1}{x}$ as x^1 rather than x^{-1}. Choice C is incorrect and may result from taking the square root of the variables in the numerator twice instead of once.